Yi Wang
College of Engineering, Lishui University
Lishui 323000, Zhejiang, China
E-mail: jefflsxy@lsu.edu.cn

June 20, 2026

To: Editor-in-Chief
IEEE Transactions on Instrumentation and Measurement

Subject: Submission of original research manuscript — "BearMamba-3: Mamba-3 State-Space Bearing Fault Diagnosis with Kinematic Frequency Inductive Bias and Multi-Sensor Fusion"

Dear Editor,

We are pleased to submit the manuscript entitled **"BearMamba-3: Mamba-3 State-Space Bearing Fault Diagnosis with Kinematic Frequency Inductive Bias and Multi-Sensor Fusion"** for consideration as a regular paper in *IEEE Transactions on Instrumentation and Measurement*. The work is original, has not been previously published, and is not currently under consideration elsewhere. All authors have approved the submission.

Why IEEE TIM. Reliable on-line vibration measurement and condition-monitoring instrumentation lies at the heart of modern predictive maintenance, and the diagnostic algorithm running on top of the measurement chain must be both physically interpretable and robust to changing operating conditions. This work proposes a measurement-stage building block—an interpretable state-space diagnostic model with explicit physical frequency alignment—that fits naturally within TIM's scope of instrumentation, signal-level measurement, and condition-based diagnostic methods.

Core contributions. The paper makes three contributions:

1. **(C1) First adoption of the official Mamba-3 backbone for bearing diagnostics.** We show that Mamba-3's complex oscillatory states carry instantaneous rotation frequencies $f_{h,k}(t)$ that are absent in real-valued SSMs (Mamba-2). This is a structural enabler, not a free choice of architecture.
2. **(C2) Kinematic frequency inductive bias $\mathcal{L}_{\mathbf{kin}}$ at the activation level.** The loss aligns Mamba-3 state frequencies with analytically derived bearing fault harmonics (BPFO, BPFI, BSF, FTF) computed per sample from the shaft speed. The constraint is undefined for real-valued SSMs, structurally binding C1 and C2 into a single integrated design choice. On CWRU, $\mathcal{L}_{\text{kin}}$ reduces inter-seed variance by 14–62% across the EAAI SNR grid.
3. **(C3) A precision-sample-efficiency trade-off in multi-sensor fusion.** Dual-sensor fusion yields a statistically significant +0.70 pp gain at −4 dB on CWRU (Wilcoxon $p = 0.031$, $n = 5$) and improves inner-race cross-condition recall by +31.6 pp on XJTU-SY (5/5 seeds), while inducing negative transfer in small-sample leave-one-bearing-out folds with a single training instance per fault type. This trade-off is reported honestly with its data-constraint origin clearly stated.

Honest scope statement. We explicitly do *not* claim within-condition accuracy supremacy on CWRU: a Phase 2b BatchNorm ablation shows that 1D-CNN with BatchNorm retains a ∼10.5 pp edge over BearMamba-3 at −8 dB, and removing BatchNorm closes this gap to within statistical noise ($p = 0.125$). The paper's distinctive contribution is in *out-of-distribution* robustness: under variable-speed cross-condition evaluation on XJTU-SY, BearMamba-3 with

$\mathcal{L}_{\text{kin}}$ achieves macro-F1 of 79.7%, +38.4 pp above 1D-CNN, demonstrating that physical inductive bias provides a form of robustness that statistical normalisation cannot replicate.

Concurrent related work. A concurrent preprint by Li et al. (arXiv:2601.21293, "Reliability-Calibrated Edge-IoT Early Fault Warning for Rotating Machinery with a Physics-Guided Tiny-Mamba Transformer", submitted to *IEEE Internet of Things Journal*) explores physics-guided Mamba for edge-IoT deployment and reliability calibration via extreme value theory. Our work and theirs occupy complementary niches—we focus on activation-level physical embedding and OOD robustness in industrial-PC deployments, they focus on sub-1 MB edge inference with EVT-calibrated false-alarm control. We discuss this relationship explicitly in Sections II and V of the manuscript.

Reproducibility. All experiments are reported with five independent random seeds, mean ± standard deviation, Wilcoxon signed-rank significance, and exhaustive hyperparameter logging. Negative and boundary-condition results ($\mathcal{L}_{\text{kin}}$ inefficacy on Paderborn discrete-speed and XJTU-SY end-of-life regimes; LOBO outer-race degradation under single training bearings) are reported alongside positive findings. Code and configurations will be released upon acceptance.

Author and conflict-of-interest declaration. Both authors contributed substantially to the conception, experimental design, analysis, and writing. We declare no conflicts of interest. The work was supported by the Natural Science Foundation of Fujian Province under Grant 2026J0011040.

Suggested reviewers. We would be glad to suggest the following experts whose expertise matches the manuscript's scope, while leaving the final selection to your discretion: (i) prof. *[Suggested Reviewer 1, name + affiliation + email]*; (ii) prof. *[Suggested Reviewer 2]*; (iii) prof. *[Suggested Reviewer 3]*. We have no objection to any reviewer chosen by the editorial board.

We hope the work will be of interest to the readership of *IEEE Transactions on Instrumentation and Measurement*, and we look forward to your editorial decision.

Yours sincerely,

Yi Wang (corresponding author)
College of Engineering, Lishui University
Lishui 323000, Zhejiang, China
jefflsxy@lsu.edu.cn

on behalf of co-author **Yongqing Tang**
School of Physics, Mechanics & Electronics, LongYan University
Longyan, Fujian, China
yq_tang@lyun.edu.cn